

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458862.65052 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1057 +/- 0.0067
Transit depth (Rp/Rs)^2	1.12 +/- 0.14 [%]
Orbital Inclination (inc)	76.97 +/- 1.06
Airmass coefficient 1 (a1)	1.017 +/- 0.0055
Airmass coefficient 2 (a2)	-0.0173 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	0.54 %
Best Comparison Star	#1 - [226, 216]
Optimal Aperture	3.22
Optimal Annulus	22.87
Transit Duration (day)	0.103 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1057 $\pm$ 0.0067 vs 0.1178 (deviation 1.81 σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 4.15 σ)
Mid-time O-C	1.2 min
Depth SNR	15.8
Residual scatter	0.54 %

Light curve

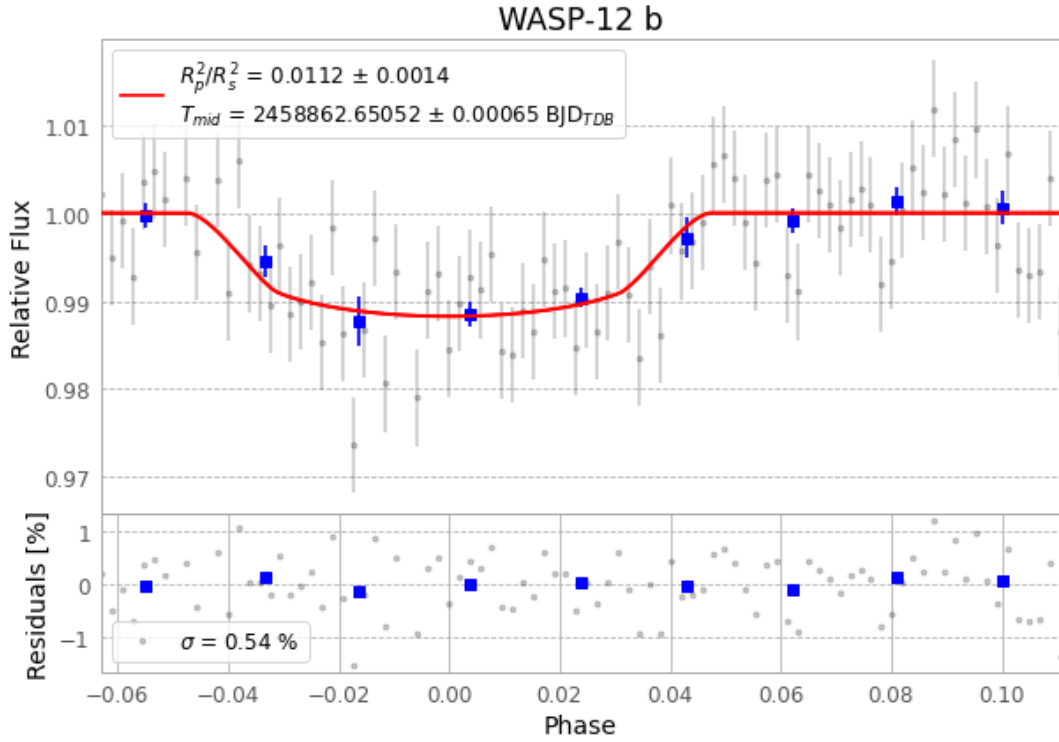


Figure 1 — FinalLightCurve_WASP-12 b_2020-01-13.png (SHA-256 161cf0b17000ff01...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 adce5ca8a69e9ab8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

92 FITS frames (60 MB), WASP-12200114014812.FITS → WASP-12200114062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200114.log (215f70a20d75...), FinalLightCurve_WASP-12 b_2020-01-13.png (161cf0b17000...), FinalLightCurve_WASP-12 b_2020-01-13.csv (d816983dbf4b...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 03:06Z.